

|                                                         |                                                                   |
|---------------------------------------------------------|-------------------------------------------------------------------|
| Program : <b>Diploma in Robotics Process Automation</b> |                                                                   |
| Course Code : <b>3308</b>                               | Course Title: <b>Digital Design and Computer Architecture Lab</b> |
| Semester : <b>3</b>                                     | Credits: <b>No Credit</b>                                         |
| Course Category: <b>Program Core</b>                    |                                                                   |
| Periods per week: <b>3 (L:0 T:0 P:3)</b>                | Periods per semester: <b>45</b>                                   |

### Course Objectives:

- To study the working of standard digital ICs and basic building blocks
- To design and implement combinational circuits
- To design and implement sequential circuits

### Course Prerequisites:

| Topic                                                                                                                     | Course Code | Course Name                                              | Semester |
|---------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------|----------|
| Basic knowledge of electronic components and equipments.                                                                  | 2039        | Fundamentals of Electrical & Electronics Engineering Lab | 2        |
| Basic functions and features of Computer, Operating system and Internet applications, basic programming skills in Python. | 1008        | Introduction to IT systems Lab                           | 1        |

### Course Outcomes:

On completion of the course, the student will be able to:

| COn        | Description                                             | Duration (Hours) | Cognitive Level |
|------------|---------------------------------------------------------|------------------|-----------------|
| <b>CO1</b> | Demonstrate functioning of various combination circuits | 12               | Applying        |
| <b>CO2</b> | Demonstrate functioning of various sequential circuits  | 12               | Applying        |
| <b>CO3</b> | To identify System Layout                               | 9                | Applying        |
| <b>CO4</b> | To install and configure DVD writer and printer         | 9                | Applying        |
|            | Series Test                                             | 3                |                 |

**CO – PO Mapping**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| <b>CO1</b>      | 3   |     | 3   | 3   |     |     |     |
| <b>CO2</b>      | 3   | 3   | 3   | 3   |     |     |     |
| <b>CO3</b>      | 3   |     |     |     |     |     |     |
| <b>CO4</b>      | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes | Name of Experiment                                                                                                                                                    | Duration (Hours) | Cognitive Level |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>      | <b>Demonstrate functioning of various combination circuits</b>                                                                                                        |                  |                 |
| M1.01           | Familiarization of Logic ICs for AND, OR, NOT, NAND, NOR and XOR by verification of truth tables and implementation of basic gates using Universal Gates (NAND & NOR) | 3                | Understanding   |
| M1.02           | Half and full adder circuits                                                                                                                                          | 3                | Applying        |
| M1.03           | Half and full subtractor circuits                                                                                                                                     | 3                | Applying        |
| M1.04           | 4: 1 MUX and 1 : 4 Demux using NAND Gates                                                                                                                             | 3                | Applying        |
| <b>CO2</b>      | <b>Demonstrate functioning of various sequential circuits</b>                                                                                                         |                  |                 |
| M2.01           | SR, J K ,D, and T Flip flops using NAND gate                                                                                                                          | 4                | Understanding   |
| M2.02           | Asynchronous mod-10 and mod 12 counter using flip-flops                                                                                                               | 4                | Applying        |
| M1.03           | Synchronous mod-10 counter using flip-flops                                                                                                                           | 4                | Applying        |
|                 | Series Test 1                                                                                                                                                         | 1.5              |                 |

|            |                                                                                                                                                                                                                                                                                              |     |               |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------|
| <b>CO3</b> | <b>To identify System Layout</b>                                                                                                                                                                                                                                                             |     |               |
| M3.01      | a) Front panel indicators & switches and Front side & rear side connectors. b) Familiarize the computer system Layout: Marking positions of SMPS, Motherboard, FDD, HDD, CD, DVD and add on cards. c) Configure BIOS setup program and troubleshoot the typical problems using BIOS utility. | 4   | Understanding |
| M3.02      | a) Install Hard Disk. b) Configure CMOS-Setup. c) Partition and Format Hard Disk. d) Identify Master /Slave / IDE Devices. e) Practice with scan disk, disk cleanup, disk De-fragmenter, Virus Detecting and Rectifying Software.                                                            | 5   | Applying      |
| <b>CO4</b> | <b>To install and configure DVD writer and printer</b>                                                                                                                                                                                                                                       |     |               |
| M4.01      | a) Install and Configure a DVD Writer and a Blu-ray Disc writer. b) Recording a Blank DVD and Blu-ray Disc.                                                                                                                                                                                  | 4   | Applying      |
| M4.02      | Printer Installation and Servicing: a) Head Cleaning in dot matrix printer b) Install and configure Dot matrix printer and Laser printer c) Troubleshoot the above printers. d) Check and connect the data cable connectivity                                                                | 5   | Applying      |
|            | Series Test 2                                                                                                                                                                                                                                                                                | 1.5 |               |

**Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                       |
|------------|------------------------------------------------|
| T1         | Electronics Lab manual Volume I, K. A. Navas , |

**Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://de-iitr.vlabs.ac.in/">https://de-iitr.vlabs.ac.in/</a>                                                               |
| 2            | <a href="http://vlabs.iitb.ac.in/vlabs-dev/labs/digital-electronics/">http://vlabs.iitb.ac.in/vlabs-dev/labs/digital-electronics/</a> |

## **\*\* - Suggested Open Ended Projects**

(Students have to do open ended experiments during the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record.)

Example:

## **Sample Questions To Test Outcomes**

1. Implementation of basic gates using Universal Gates.
2. Implement half adder and full adder
3. Implement half subtractor and full subtractor
4. Implement 4:1 Mux and 1:4 Demux using NAND gates
5. Implement S R, T, D and J K flipflop using NAND gates
6. Set up mod 11 asynchronous counter
7. Set up a mod 10 synchronous counter
8. Install a Laser printer and print the test page.
9. Install a DVD writer and configure it in slave mode.
10. Configure the IP address of a network enabled printer.
11. Refill the Toner Cartridge of a Laser printer.
12. Clean the lens of a DVD player and lubricate the mechanism.